

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1583

John P Madgett 100 kV
April 26, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1583. The Interconnection Customer proposes a 172.0 MW wind facility interconnecting at the John P Madgett 100 kV substation in the Ohio area. The request was received on October 02, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 476 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$17,719,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Getty Substation to Csumnes 100 KV
Contingency: Getty to Ix Ctr 100 KV
Loading: 149.9% (Rating: 153 MVA)
DFAX: 6.7%
Pre-project loading: 142.3%
2. Facility: Fowles to Cosumnes 100 KV L
Contingency: Getty to Ix Ctr 100 KV
Loading: 149.7% (Rating: 133 MVA)
DFAX: 6.7%
Pre-project loading: 141.1%
3. Facility: West Union to Loring Power Plant 100 KV
Contingency: Imperial to Eufaula 500 KV
Loading: 145.1% (Rating: 118 MVA)
DFAX: 6.4%
Pre-project loading: 135.8%
4. Facility: Beach Grove to Dollarhide 100 KV
Contingency: Eufaula to Beaver 500 KV
Loading: 142.8% (Rating: 133 MVA)
DFAX: 46.5%
Pre-project loading: 82.9%
5. Facility: Fox to Cedar Hill 100 KV L
Contingency: Imperial to Eufaula 500 KV
Loading: 136.2% (Rating: 104 MVA)
DFAX: 5.4%
Pre-project loading: 127.3%

6. Facility: Ix Ctr to Getty 100 KV L
Contingency: Eufaula to Imperial Substation 500 KV line
Loading: 118.5% (Rating: 241 MVA)
DFAX: 7.0%
Pre-project loading: 113.5%
7. Facility: Velasco to Cedar Hill 100 KV
Contingency: Eufaula to Imperial 500 KV
Loading: 112.1% (Rating: 153 MVA)
DFAX: 5.4%
Pre-project loading: 106.1%
8. Facility: Ix Center to Velasco 100 KV
Contingency: Imperial to Eufaula 500 KV
Loading: 109.7% (Rating: 169 MVA)
DFAX: 5.4%
Pre-project loading: 104.2%
9. Facility: John P Madgett to Dollarhide 100 KV
Contingency: Beavr to Eufaula 500 KV
Loading: 109.2% (Rating: 299 MVA)
DFAX: 59.7%
Pre-project loading: 74.9%
10. Facility: Don Julio-Lorac Vista to Gallatin Tn 100 KV
Contingency: Beaver to Eufaula 500 KV
Loading: 102.1% (Rating: 120 MVA)
DFAX: 8.0%
Pre-project loading: 90.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Getty Substation to Csumnes 100 KV	Rebuild line with higher-capacity conductor	\$1,504,000	100% (\$1,504,000)
Fowles to Cosumnes 100 KV L	Rebuild line with higher-capacity conductor	\$3,504,000	100% (\$3,504,000)
West Union to Loring Power Plant 100 KV	Rebuild line with higher-capacity conductor	\$1,247,000	100% (\$1,247,000)
Beach Grove to Dollarhide 100 KV	Rebuild line with higher-capacity conductor	\$1,844,000	100% (\$1,844,000)
Cedar Hill to Fox 100 KV	Rebuild line with higher-capacity conductor	\$1,679,000	100% (\$1,679,000)
Ix Ctr to Getty 100 KV L	Rebuild line with higher-capacity conductor	\$2,373,000	100% (\$2,373,000)
Velasco to Cedar Hill 100 KV	Rebuild line with higher-capacity conductor	\$3,056,000	100% (\$3,056,000)
Ix Ctr to Velaco 100 KV Line	Reconductor line	\$692,000	100% (\$692,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
John P Madgett to Dollarhide 100 KV	Reconductor line	\$720,000	84% (\$602,000)
Gallatin Tn to Don Julio-Lorac Vista 100 KV	Reconductor line	\$2,286,000	53% (\$1,218,000)
	Total allocated to project		\$17,719,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.